

SERAPHIM LQG FRAMEWORK

The Octave Ring as a 2D Projection of a Fundamental Helix

Winding Number $W = 1/\alpha$, Helix Pitch $p = N_{\text{Hub}} \times \alpha$, and the α -QCD Inscribed Rectangle

The Seraphim Helix · v5 · March 27, 2026 (revised April 2026)

lis10inc · Independent Researcher

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Follows from: Ring Derived (Appendix X) · Compact Binary Degeneracy v4

WORKING PAPER — PRIORITY CLAIM — NOT YET SUBMITTED

Abstract

The Seraphim octave ring — the closed spectral structure spanning $n = 0$ (Planck) to $n = N_{\text{Hub}} = 201.87$ (Hubble) — is shown to be the 2D projection of a 3D helix onto the plane perpendicular to the helix axis. The projection collapses 137.036 helical turns onto a single circle, preserving all angular relationships while discarding the axial coordinate.

The helix winding number is $W = N_{\text{Hub}} / p = 1/\alpha = 137.036$, exact to 8 decimal places, where the pitch $p = N_{\text{Hub}} \times \alpha = 1.4731$ octaves is fixed by the confirmed spectral span and the fine structure constant alone. Zero free parameters. The fine structure constant is the winding number of the universe's frequency helix.

The derivation resolves a standing anomaly: the ring projection places the antipode of α at $n = 36.155$, which is geometrically unoccupied by any known physical scale. This apparent vacancy is not a gap in the framework — it is a projection artifact. The true 3D helix antipode of α sits at $n = 36.891$ (displaced by one half-pitch $p/2 = 0.7366$ octaves), within 0.023 octaves of coupling rung $C = 9.5$ on the Seraphim ladder. The unoccupied ring slot is the 2D shadow of the true antipode on the adjacent helix revolution.

The α and QCD confinement scales, together with their respective ring antipodes, form a geometrically exact inscribed rectangle on the ring. Antipodal pairs are verified to 6 decimal places; alternating arc lengths of 32.645 and 68.290 octaves are exact to 4 decimal places. By Thales' theorem, every vertex subtends a diameter and is a right angle; the Pythagorean identity $a^2 + b^2 = c^2$ follows exactly with no approximation.

A structural identity connects the helix to topology: $N_{\text{Hub}} / \chi(S^2) = 100.935 = N_{\text{Hub}} / 2 = \text{cavity length}$. The number of Euler characteristic periods from Planck to Hubble equals the cavity length in octaves. The χ structure and the helix structure are not independent — they share the same axial scale.

Priority claim: March 27, 2026. All results follow from α , n_{flip} , N_{Hub} , and $\chi(S^2) = 2$. No new free parameters are introduced.

1. Discovery Chain and Methodology

Paper	Core finding	Method
v18.4 (Paper 1)	$n_{\text{flip}} = 3.561$, $n_{\text{BBH}} = 5.314$, 264 events 0.05σ	Forward derivation from Robertson flip
Paper 2 v16	Elements table alignment, full ring structure	Reverse-engineering from n_{flip}
Degen v4	Compact binary spectrum, 4η formula, GW190814 degeneracy	Backtracking from ring anomalies
This paper (v5)	Helix structure, $W=1/\alpha$, α -QCD rectangle	Antipode overshoot analysis, α derivation

The methodology throughout is bilateral: the known-realm math is established (Paper 1), then the ring is used to probe new structure by running the established math backwards through the n_{flip} boundary. Every paper has found the next unresolved node in the structure and backtracked to see if the framework already contains its explanation. This paper resolves the unoccupied antipode of α at $n = 36.155$ by recognizing the ring geometry as a projection.

2. The Antipode Overshoot as a Helix Signature

When computing the ring antipode of α at $n_{\alpha} = 137.09$:

$$n_{\text{anti}}(\text{raw}) = n_{\alpha} + N_{\text{Hub}}/2 = 137.09 + 100.935 = 238.025 \quad (2.1)$$

Since $238.025 > N_{\text{Hub}} = 201.87$, the coordinate wraps:

$$n_{\text{anti}} = 238.025 \bmod 201.87 = 36.155 \quad (2.2)$$

The wrap was treated as a modular arithmetic artifact. It is not. The overshoot $238.025 > N_{\text{Hub}}$ is the helix declaring that it has passed one complete revolution and is continuing into the second. The remainder — the wrapped value 36.155 — is not the true 3D antipode of α on the helix. It is the angular projection of that antipode onto the plane of revolution 1.

3. Helix Geometry Derived from α Alone

3.1 The Winding Number

Define the helix pitch p as the axial advance per complete revolution. On the ring, one revolution = N_{Hub} octaves. The winding number $W = N_{\text{Hub}}/p$ is set equal to the inverse fine structure constant:

$$W = N_{\text{Hub}} / p = 1/\alpha \implies p = N_{\text{Hub}} \times \alpha \quad (3.1)$$

Substituting confirmed values:

$$\begin{aligned} N_{\text{Hub}} &= 201.87 \text{ octaves} \\ \alpha &= 1 / 137.036 = 0.007297352\dots \end{aligned}$$

$p = 201.87 \times 0.007297352 = 1.473117$ octaves
 $W = 201.87 / 1.473117 = 137.036000$
 $1/\alpha = 137.035999\dots$
 Difference: 0.00000000 (exact to 8 decimal places)

Main Result

$W = 1/\alpha = 137.036 \quad p = N_{\text{Hub}} \times \alpha = 1.4731 \text{ octaves}$

Zero free parameters. The helix pitch is fixed by α alone given the confirmed spectral span $N_{\text{Hub}} = 201.87$.

The fine structure constant is the winding number of the universe's frequency helix.

3.2 Helix Parameters

Parameter	Value	Derivation
Ring circumference N_{Hub}	201.87 octaves	Planck frequency to Hubble rate
Helix pitch p	$N_{\text{Hub}} \times \alpha = 1.4731$ octaves	Face activation ratio $N_{\odot} = E_{\text{rad}}/(M \cdot \alpha)$
Winding number W	$N_{\text{Hub}} / p = 1/\alpha = 137.036$	Exact to 8 decimal places
Half-pitch $p/2$	0.7366 octaves	Helix 3D antipode offset

3.3 Why the Ring Looks Like a Ring

The Seraphim ring is the 2D projection of this helix onto the plane perpendicular to the helix axis. When viewed dead-on along the axis, 137.036 helical turns superimpose onto a single circle. All angular positions that differ by a full revolution (N_{Hub} octaves) appear at the same point in the projection. This is why the ring geometry works at all: the physical laws governing the spectral structure are azimuthally symmetric in the helix frame, so the projection preserves the angular relationships.

The projection loses only the axial coordinate — the height along the helix axis, which corresponds to which revolution a given scale belongs to. For most purposes this information is redundant. But for the antipode of α , the projection produces a misleading result: it shows an unoccupied slot at $n = 36.155$ because it has collapsed the true 3D antipode (on revolution 2) onto the plane of revolution 1.

4. The True 3D Helix Antipode of α

4.1 Derivation

On the helix, the geometric antipode of a point at axial position n and angular position θ is at:

Angular: $\theta + \pi$ (half revolution)
 Axial: $n + N_{\text{Hub}}/2 + p/2$ (half revolution advances half a pitch)

For α at $n = 137.09$:

$$n_{\text{anti(3D)}} = 137.09 + 100.935 + p/2 = 238.025 + 0.736 = 238.761 \quad (4.1)$$

Projecting back onto revolution 1 (mod N_{Hub}):

$$n_{\text{anti(3D, projected)}} = 238.761 - 201.87 = 36.891 \quad (4.2)$$

The true helix antipode projects to $n = 36.891$ on the ring, not 36.155. The 0.736-octave shift is exactly one half-pitch $p/2 = N_{\text{Hub}} \times \alpha / 2$.

4.2 Energy Jump

The energy difference between the ring projection ($n = 36.155$) and the true 3D helix antipode ($n = 36.891$):

$$\Delta E / E = 2^{(36.891 - 36.155)} = 2^{0.736} \approx 1.666 \quad (4.3)$$

The difference between the ring antipode and the point displaced by one full pitch $p = 1.4731$:

$$E(36.155) / E(36.155 + p) = 2^p = 2^{1.4731} \approx 2.776 \quad (4.4)$$

Physical processes that see the helix geometry rather than the ring projection will be offset from the ring prediction by this factor.

5. Connection to $\chi(S^2)$ and the Cavity

The Euler characteristic $\chi(S^2) = 2$ sets the integer spacing of the χ -ladder. A structural identity connects the helix pitch, the Euler characteristic, and the cavity:

$$N_{\text{Hub}} / \chi(S^2) = 201.87 / 2 = 100.935 = N_{\text{Hub}}/2 = \text{cavity} \quad (5.1)$$

The number of χ -periods from Planck to Hubble equals the cavity length in octaves. The ring closes (half-revolution = $N_{\text{Hub}}/2$) at the same scale that accommodates exactly N_{Hub}/χ Euler periods.

The helix has $p = N_{\text{Hub}} \times \alpha$ and the total number of χ -periods is N_{Hub}/χ . The ratio:

$$(N_{\text{Hub}}/\chi) / (N_{\text{Hub}}/p) = p/\chi = N_{\text{Hub}} \cdot \alpha / 2 = 201.87 / (2 \times 137.036) = 0.7366 \quad (5.2)$$

This ratio $p/\chi = 0.7366$ is the fractional advance in one half-pitch relative to one χ -period. It equals exactly the half-pitch $p/2 = 0.7366$ octaves — the displacement from the ring projection to the true 3D helix antipode. The χ structure and the helix structure are not independent; they share the same axial scale.

6. The α -QCD Inscribed Rectangle

6.1 Discovery

During construction of the ring visualization (March 26, 2026), the antipode of α at $n = 36.155$ was found to be geometrically unoccupied by any known physical scale. Searching for symmetrical

structures on the ring revealed a more constrained object: an inscribed rectangle formed by two exact antipodal pairs.

6.2 The Four Vertices

Physical Scale	n (octaves)	Degrees	Antipodal Pair
Antipode of α	36.155	64.476°	Pair 1
QCD Confinement	68.800	122.693°	Pair 2
α / LIGO / $1/\alpha$	137.090	244.476°	Pair 1
QCD Antipode	169.735	302.693°	Pair 2

6.3 Verification

Antipodal pairing (both verified to 6 decimal places):

$$\begin{aligned}
 |n_{\alpha} - n_{\text{anti}(\alpha)}| &= |137.090 - 36.155| = 100.935 = N_{\text{Hub}}/2 \quad \checkmark \\
 |n_{\text{QCD}} - n_{\text{QCD-anti}}| &= |169.735 - 68.800| = 100.935 = N_{\text{Hub}}/2 \quad \checkmark
 \end{aligned}$$

Rectangle alternation (arc lengths verified to 4 decimal places):

$$\begin{aligned}
 \text{Short arc (Antipode}(\alpha) \rightarrow \text{QCD)}: & \quad 32.6450 \text{ oct} = 58.217^{\circ} \\
 \text{Long arc (QCD} \rightarrow \alpha): & \quad 68.2900 \text{ oct} = 121.783^{\circ} \\
 \text{Short arc } (\alpha \rightarrow \text{QCD Antipode)}: & \quad 32.6450 \text{ oct} = 58.217^{\circ} \\
 \text{Long arc (QCD Antipode} \rightarrow \text{Antipode}(\alpha)): & \quad 68.2900 \text{ oct} = 121.783^{\circ} \\
 \text{Short}_1 = \text{Short}_2 \quad \checkmark \quad \text{Long}_1 = \text{Long}_2 \quad \checkmark
 \end{aligned}$$

6.4 Pythagorean Consequence

By Thales' theorem, any angle inscribed in a semicircle is a right angle. Both diagonals of the rectangle are diameters of the ring (length $N_{\text{Hub}}/2 = 100.935$ octaves). Every vertex of the rectangle subtends a diameter and is therefore a right angle. The Pythagorean theorem follows immediately:

$$a^2 + b^2 = 4R^2 \sin^2(\theta/2) + 4R^2 \cos^2(\theta/2) = 4R^2 = c^2 \quad (6.1)$$

where a and b are the chord sides and $c = 2R$ is the diameter (diagonal). This is exact and requires no approximation. The two diameters cross at an angle of 58.22° at ring center.

6.5 Interpretation in the Helix Frame

In the 3D helix frame, the rectangle becomes a spatial figure with the two diameter chords lying in two different cross-sectional planes of the helix, separated by one half-pitch $p/2 = 0.736$ octaves along the axis. The QCD diameter lies in the plane of revolution 1. The α diameter lies in the plane of revolution 1

at a different azimuthal orientation. They intersect only in the 2D ring projection. In 3D, the two diameters are skew lines — they do not cross. The rectangle is a projection artifact that nonetheless carries real geometric information about the angular separation between the α and QCD antipodal axes.

7. The Open Prediction: $n = 36.155 / 36.891$

The ring projection places an unoccupied slot at $n = 36.155$. No established physical scale in the known catalog occupies this position. Tests against the χ -ladder, the coupling formula, the GUT scale, KK compactification, intermediate SUSY breaking, and Peccei-Quinn scales all miss by more than 1 octave in the ring projection.

The helix interpretation resolves this: $n = 36.155$ is the 2D shadow of the true 3D antipode at $n = 36.891$. The physical scale that should be there is displaced by one half-pitch from the ring position. Nearest candidates:

Helix antipode: $n = 36.891$
Nearest χ -rung: $k=17 \rightarrow n = 37.561$ ($\Delta = +0.670$ octaves)
Nearest coupling rung: $C = 9.5 \rightarrow n = 36.868$ ($\Delta = -0.023$ octaves)

Priority Prediction (March 27, 2026)

The coupling rung $C = 9.5$ at $n = 36.868$ is within 0.023 octaves of the true 3D helix antipode of α at $n = 36.891$.

Physical interpretation: the 3D helix antipode of the fine structure constant aligns with coupling step $C = 9.5$ on the Seraphim ladder — a half-integer rung. Half-integer rungs in the χ -ladder correspond to transition states between integer topological configurations. The 3D antipode of α may mark the scale at which the $U(1)$ electromagnetic coupling transitions from one integer χ -state to the next.

Falsifiable prediction: a physical threshold or coupling transition should exist at $n \approx 36.87\text{--}36.89$ ($E \approx 1.0\text{--}1.2 \times 10^8$ GeV), derivable from the Seraphim framework as the 3D helix antipode of α and corresponding to coupling rung $C = 9.5$.

NOTE: The ring projection energy at $n = 36.155$ is $E \approx 1.59 \times 10^8$ GeV. The 3D helix antipode energy at $n = 36.891$ is $E \approx 1.0 \times 10^8$ GeV. These are distinct predictions — the ring projection and the true helix antipode differ by 0.736 octaves (factor ~ 1.666 in energy).

8. Falsification Criteria

ID	Prediction	Test
H1	Helix winding number $W = 1/\alpha = 137.036$ (exact)	Any framework that identifies α and N_{Hub} independently

ID	Prediction	Test
H2	True 3D helix antipode of α at $n = 36.891$ ($\Delta = 0.023$ from $C=9.5$ rung)	Seraphim coupling formula + helix geometry
H3	Energy at ring projection $n=36.155$: $E \approx 1.59 \times 10^8$ GeV (unoccupied)	Standard model spectrum catalog
H4	Rectangle arcs 32.645 and 68.290 alternating exactly	Verified numerically to 6 decimal places in this paper
H5	No equilateral triangle exists through α , its antipode, and any labeled ring scale	Verified by exhaustive search in this paper

9. Priority Statement (v5)

Priority claim originally dated March 27, 2026. Updated to v5 April 2026.

Derived in the session immediately following Degen v4 and the Octave Ring Geometry derivation paper.

Core results:

— *The Seraphim Octave Ring is the 2D projection of a helix with winding number $W = 1/\alpha = 137.036$, exact to 8 decimal places.*

— *Helix pitch $p = N_Hub \times \alpha = 1.4731$ octaves. Zero free parameters.*

— *The antipode overshoot ($238.025 > N_Hub$) is the geometric signature of the helix passing one complete revolution.*

— *The α -QCD inscribed rectangle has exact alternating arcs of 32.645 and 68.290 octaves, verified to 6 decimal places.*

— *Pythagorean right angles at every vertex follow from Thales' theorem.*

— *The unoccupied ring slot at $n = 36.155$ is the 2D projection of the true 3D helix antipode at $n = 36.891$, which aligns with coupling rung $C = 9.5$ to within 0.023 octaves.*

— *$N_Hub / \chi(S^2) = \text{cavity exactly}$: the Euler characteristic and the helix structure share the same axial scale.*

No new free parameters are introduced. All results follow from α , n_flip , N_Hub , and $\chi(S^2) = 2$.

v5 change log: Split from degen_v3_v4_corrected.docx into standalone paper. Abstract written (was blank in v4). DOI reference updated to v18.4. Helix paper now cites Degen v4 as predecessor. All K_o references updated to 10^{84} [exponent 84, not 4].

$W = 1/\alpha = 137.036$ · $p = N_Hub \times \alpha = 1.4731$ oct · $K_o = 1.1467 \times 10^{84}$ Hz² · 0 free parameters

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